



**Progressive Education Society's
Modern College of Engineering, Pune-05.**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Department of S.Y. B. Tech [AIDS]

Assignment No: 01

SUBJECT: Computer Networks

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Class: SY BTech (AIDS)

Section: A

Roll Number: 26048

Title / Problem Statement:

Use Windows/Ubuntu operating system for basic Network Commands Practice.

Input/Output :

1) ipconfig

```
Microsoft Windows [Version 10.0.20190.6595]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : pccmodecoms.edu.in
    Link-local IPv6 Address . . . . . : fe80::6931:b6a8:bd38:6e1a%12
    IPv4 Address. . . . . : 172.16.12.91
    Subnet Mask . . . . . : 255.255.250.0
    Default Gateway . . . . . : 192.168.16.1

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::f43b:6b6f:9043:723e%4
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.250.0
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Ethernet adapter vEthernet (WSL (Hyper-V Firewall)):

    Connection-specific DNS Suffix . :
```

2) ping

```
Microsoft Windows [Version 10.0.20190.6595]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
           [-r count] [-w count] [[-j host-list] | [-k host-list]]
           [-s timeout] [-R] [-S srcaddr] [-c compartment] [-p]
           [-u] [-6] target_name

Options:
  -t           Ping the specified host until stopped.
               To see statistics and continue - type Control-Break;
               To stop - type Control-C.
  -a           Resolve addresses to hostnames.
  -n count     Number of echo requests to send.
  -l size      Send buffer size.
  -f           Set Don't Fragment flag in packet (IPv4-only).
  -i TTL       Time To Live.
  -v TOS       Type of Service (IPv4-only. This setting has been deprecated
               and has no effect on the type of service field in the IP
               header).
  -r count     Record route for count hops (IPv4-only).
  -j host-list  Loose source route along host-list (IPv4-only).
  -k host-list  Strict source route along host-list (IPv4-only).
  -w timeout   Timeout in milliseconds to wait for each reply.
  -S srcaddr   Use routing header to test reverse route also (IPv6-only).
               Per RFC 6095 the use of this routing header has been
               deprecated. Some systems may drop echo requests if
               this header is used.
  -S srcaddr   Source address to use.
  -c compartment Routing compartment identifier.
  -p           Ping a Hyper-V Network Virtualization provider address.
  -u           Force using UDP.
  -6           Force using IPv6.
```

3) netstat

```

Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:49675          kubernetes:49674       ESTABLISHED
TCP    127.0.0.1:49676          kubernetes:49673       ESTABLISHED
TCP    127.0.0.1:49676          kubernetes:49676       ESTABLISHED
TCP    172.16.12.91:68668      sc-16-F100:5228       ESTABLISHED
TCP    172.16.12.91:68623      ps-16-F04:844ps       ESTABLISHED
TCP    172.16.12.91:59617      88.225.239.96:8080    ESTABLISHED
TCP    172.16.12.91:68627      168:844ps             ESTABLISHED
TCP    172.16.12.91:68668      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68669      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68618      52.113.196.256:https  ESTABLISHED
TCP    172.16.12.91:68611      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68612      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68613      4.168.268.268:https   ESTABLISHED
TCP    172.16.12.91:68614      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68616      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68616      131.253.33.258:https  ESTABLISHED
TCP    172.16.12.91:68619      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68618      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68619      204.79.197.222:https  ESTABLISHED
TCP    172.16.12.91:68620      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68621      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68622      a23-58-10-13:844ps   ESTABLISHED
TCP    172.16.12.91:68624      184.18.23.222:https   ESTABLISHED
TCP    172.16.12.91:68626      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68626      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68627      13.187.138.286:https  ESTABLISHED
TCP    172.16.12.91:68628      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68629      PDC:domain            TIME_WAIT
TCP    172.16.12.91:68630      52.123.128.256:https  ESTABLISHED
TCP    172.16.12.91:68631      PDC:domain            TIME_WAIT

```

4) tracert

```

Microsoft Windows [Version 10.0.26100.6584]
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C:\Users\admin>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d          Do not resolve addresses to hostnames.
    -h maximum_hops  Maximum number of hops to search for target.
    -j host-list  Loose source route along host-list (IPv4-only).
    -w timeout    Wait timeout milliseconds for each reply.
    -R          Trace round-trip path (IPv6-only).
    -S srcaddr    Source address to use (IPv6-only).
    -4          Force using IPv4.
    -6          Force using IPv6.

```

5) ns lookup

```

C:\Users\admin>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

-a          Displays current ARP entries by interrogating the current
            protocol data. If inet_addr is specified, the IP and Physical
            addresses for only the specified computer are displayed. If
            more than one network interface uses ARP, entries for each ARP
            table are displayed.
-g          Same as -a.
-v          Displays current ARP entries in verbose mode. All invalid
            entries and entries on the loop-back interface will be shown.
inet_addr   Specifies an internet address.
-N if_addr  Displays the ARP entries for the network interface specified
            by if_addr.
-d          Deletes the host specified by inet_addr. inet_addr may be
            wildcarded with * to delete all hosts.
-s          Adds the host and associates the Internet address inet_addr
            with the Physical address eth_addr. The Physical address is
            given as 6 hexadecimal bytes separated by hyphens. The entry
            is permanent.
eth_addr    Specifies a physical address.
if_addr     If present, this specifies the Internet address of the
            interface whose address translation table should be modified.
            If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
> arp -a                                     .... Displays the arp table.

C:\Users\admin>

```

6)route

```
C:\Users\admin>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                                [MASK netmask] [gateway] [METRIC metric] [IF interface]

-f          Clears the routing tables of all gateway entries.  If this is
            used in conjunction with one of the commands, the tables are
            cleared prior to running the command.

-p          When used with the ADD command, makes a route persistent across
            boots of the system. By default, routes are not preserved
            when the system is restarted. Ignored for all other commands,
            which always affect the appropriate persistent routes.

-4          Force using IPv4.

-6          Force using IPv6.

command     One of these:
            PRINT      Prints a route
            ADD        Adds a route
            DELETE     Deletes a route
            CHANGE     Modifies an existing route

destination Specifies the host.
MASK          Specifies that the next parameter is the 'netmask' value.
netmask       Specifies a subnet mask value for this route entry.
            If not specified, it defaults to 255.255.255.255.
gateway       Specifies gateway.
interface     the interface number for the specified route.
METRIC        specifies the metric, ie. cost for the destination.
```

7) getmac

```
C:\Users\admin>getmac

Physical Address      Transport Name
=====
D8-BB-C1-EE-92-D4    \Device\Tcpip_{97B5F99B-F39D-4BD4-8086-4EF546705D98}
DC-21-48-05-41-8C    Media disconnected
0A-00-27-00-00-04    \Device\Tcpip_{26FF6F2F-A51E-4365-8D27-DE1E54419F87}
```

8) nslookup

```
C:\Users\admin>nslookup  
Default Server:  PDC.PESMODERNCOE.EDU  
Address:  192.168.11.100
```